Education and Information Technologies (2024) 29:2277–2301
https://doi.org/10.1007/s10639-023-11908-0

Identifying key features of resilient students in digital reading: Insights from a machine learning approach

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Received: 20 March 2023 / Accepted: 22 May 2023 / Published online: 1 June 2023

Abstract

With the rapid growth of education data in large-scale assessment, machine learning techniques are crucial to the interdisciplinary development of education and information. Although data mining tools are increasingly used to predict overall student performance, resilient students in the digital world remain unstudied. Our study aims to comprehensively identify key features in the classification of academically resilient students (ARS) and non-academically resilient students (NRS) in digital reading. With a sample of 11,496 disadvantaged students from seven high-performing Asian economies, data drawn from the Programme for International Student Assessment (PISA) 2018 were analyzed through Support Vector Machine (SVM). Results indicated that 20 key features were selected from 105 contextual features at the individual, home, and school levels, which demonstrated a high predictive ability of the model. Personal experience, especially the use of metacognitive strategies in digital reading and reading enjoyment were predominant features. Interestingly, information and communication technology (ICT) resources and usage showed mixed effects on resilient students. This study provides significant implications for cultivating resilient students in online learning environments.

Keywords Support vector machine · Digital reading · Academic resilience · Asia · PISA

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1 Introduction

As a branch of artificial intelligence, machine learning (ML) focuses on instructing computers to function without explicitly programmed and acquire knowledge intelligently (Fahd et al., 2022). Nowadays, ML has gradually developed its application in education, including student modeling, scientific inquiry, adaptive system, evaluation, and decision support systems (Bakhshinategh et al., 2018). ML could avoid multicollinearity issues and highly correlated effects caused by a massive group of variables, while traditional statistical methods might suffer from these problems (Senaviratna & Cooray, 2019). In recent years, the superiority of ML approach in predicting overall student performance has been demonstrated in numerous research (e.g., Batool et al., 2023; Chen & Zhai, 2023; Qazdar et al., 2019).

In the data-driven era, it is critical for students to access information through digital reading skills. Most studies investigated contextual factors of students' reading performance in an online learning environment (e.g., Naumann, 2015; Yu et al., 2023). It is commonly evidenced that disadvantaged socioeconomic status (SES) students demonstrate lower digital reading performance than their advantaged SES counterparts (e.g., Erdem & Kaya, 2021; Yeung et al., 2022). However, there is a group of low SES students who beat against the odds and acquire high levels of digital reading literacy (e.g., Jang et al., 2023). In this sense, it is crucial to identify the features of academically resilient readers in today's digital world.

Academic resilience is a phenomenon describing individuals from disadvantaged SES families still achieve academic success. (Agasisti et al., 2021; OECD, 2011). Since the understanding of this phenomenon is conducive to narrowing the achievement gaps between high and low SES groups, academic resilience has been of growing interest in the last two decades (Rudd et al., 2021; OECD, 2011). It has been extensively examined in the international large-scale assessment (ILSA) research, such as the Programme for International Student Assessment (PISA) (e.g., Agasisti et al., 2021; Cheung, 2017), the Progress in International Reading Literacy Study (PIRLS) (e.g., Cordero & Mateos-Romero, 2021; García-Crespo et al., 2019), and the Trends in International Mathematics and Science Study (TIMSS) (e.g., Aydın & Erdem, 2022; Sandoval-Hernández & Białowolski, 2016). Apart from the cross-sectional ILSA data, previous studies adopted a longitudinal lens in academic resilience research (e.g., Cappella & Weinstein, 2001; Kim et al., 2021). Moreover, the qualitative approach is commonly utilized to examine academic resilience (e.g., Kim et al., 2018; Singh, 2021; Tudor et al., 2020). However, much less attention has been paid to identifying key features of resilient students through ML techniques.

Notably, identifying factors of academic resilience is critical in two ways. First, the proportion of resilient students mirrors the degree of education equality in the system under study (Agasisti et al., 2017). Second, it plays a pivotal role in promoting the quality of educational systems (Clavel et al., 2022). Nevertheless, a significant body of previous research only offered a snapshot to examine factors of academic resilience, such as learning characteristics (Cheung, 2017; Cheung et al., 2014), parenting practices, and child-parent interactions (Li & Yeung, 2019; Ma & Wu, 2019), as well as school climate and resources (Agasisti et al., 2021; Teng, 2020). Hence, there is an urgent need to identify contextual factors from a holistic perspective. While most research utilized traditional statistical methods in search of key factors (e.g., Agasisti & Longobardi, 2014; Li & Yeung, 2019; Ma & Wu, 2019), which further brings huge challenges to adopt an integrated perspective in the ILSA research. In addition, the development of resilience is influenced by the culture that shapes the construction of meaning (Ungar, 2011). To clarify, the expression of academic resilience might vary between Western and Eastern economies (e.g., Özcan & Bulus, 2022). Since early literature views resilience as a fixed individual trait, the individualism of Western culture might be utilized to interpret resilience (Ungar, 2013). Therefore, most academic resilience research heavily emphasized Western culture at the early stage, and little attention has been paid to Asia (Li et al., 2017).

2 Literature review

2.1 Digital reading literacy

In PISA 2018, digital reading literacy is defined as "understanding, using, evaluating, reflecting on and engaging with digital texts in order to achieve one's goals, to develop one's knowledge and potential and to participate in society" (OECD, 2019a, p. 26). Compared with printed reading, digital reading contains more complicated and dynamic components such as multimedia animation and pop-up windows (Leu et al., 2013). It therefore requires students to evaluate and interpret information critically. In this regard, contributing factors of digital reading are major concerns in an online learning environment. Reading behaviors, reading dispositions, and student background are three core individual domains contributing to digital reading performance. For example, students' attitudes towards ICT, reading self-efficacy, metacognition strategies, and navigation skills positively predicted digital reading (Chen et al., 2022; Lim & Jung, 2019). Moreover, gender, SES, and educational resources at home are crucial indicators of digital reading (Chen et al., 2022; Wu, 2014; Xiao & Hu, 2019). Concerning the school domain, school characteristics and school process factors are two contributing sources of digital reading. On one side, high SES schools and schools located in urban areas showed positive impacts on digital reading (Combrinck & Mtsatse, 2019; Segers et al., 2023). On the other side, supportive disciplinary climate, teachers' stimulation of reading engagement, and adaptative instruction foster digital reading performance (Hu & Wang, 2022; Qian & Lau, 2022).

2.2 Academic resilience

For the last few decades, resilience has undergone four waves of theoretical development (Luthar, 2015). Aligned with the latest wave of resilience research, a social-ecological perspective defines resilience as “the capacity of both individuals and their environments to interact in ways that optimize developmental processes” (Ungar, 2013, p.256). It thus highlights the joint effects of key factors on resilience. Up to now, resilience has been extensively applied in various disciplines, including mental health (Pinheiro et al., 2022), psychology (Song et al., 2019), sociology (Adger, 2000), ecology (Gunderson, 2000), and education (Wang et al., 1994). For a long time, resilience research has sought to break down the barriers between social and natural sciences (Hamborg et al., 2020). Until now, the latest wave of resilience research has facilitated multidisciplinary development (e.g., genes, neuroscience, psychology) (Luthar, 2015).

Within the education domain, academic resilience refers to “the heightened likelihood of success in school and other life accomplishments, despite environmental adversities brought about by early traits, conditions, and experiences” (Wang et al., 1994, p.137). Derived from this definition, academic resilience is made up of adversity and positive adaptation (Windle, 2011). These two major components determine a definition-driven approach to operationalize academic resilience (Rudd et al., 2021). By adopting a comparable lens, our study defines academically resilient students as disadvantaged individuals who are located at the bottom 25% of economic, social, and cultural status (ESCS) within their economy but whose performance achieve at the top 25% across seven Asian economies, following the guidelines of OECD reports (OECD, 2011, 2019a).

2.3 Influential factors of academic resilience

Garmezy (1985) has proposed three dimensions that contributed to resilience, including personal features, relationships, and external supportive systems. From a micro perspective, key contextual factors were divided into three components: 1) I am (individual’s internal strengths); 2) I can (individual’s interpersonal and social skills); 3) I have (external resources) (Grotberg, 1995). Overall, key features are categorized into internal and external domains (Luthar et al., 2000). Following the PISA 2018 assessment and analytical framework (OECD, 2019b), it is suggested to classify influential factors of academic resilience into student, home, and school contexts.

2.3.1 Student context

Kumpfer (1999) has proposed a resilience framework to describe individual factors in five dimensions: cognitive, motivational, behavioral, emotional, and physical. Primarily, regarding the cognitive dimension, self-regulated learning, critical thinking, and peer learning were positively associated with academic resilience (Mohan & Verma, 2020). The second is referring to the motivational domain. Students who had high levels of self-efficacy tended to be academically resilient (Ferrera & Rodríguez, 2015; Wang et al., 2022), while anxiety was more likely to undermine academic resilience (Alivernini et al., 2016; Martin & Marsh, 2006). Moreover, high levels of learning interests and enjoyment (Clavel et al., 2022; Özcan & Bulus, 2022) contributed to academic resilience. Third, concerning behavioral engagement, school attendance yielded significantly positive effects on academic resilience (Martin et al., 2022). Instead, low levels of absenteeism and truancy were risk factors for academic resilience (Agasisti et al., 2021; Cordero & Mateos-Romero, 2021). Fourth, a stable emotional state promoted academic resilience (Tamannaeifar & Shahmirzaei, 2019). Positive future aspirations (Gizir & Aydin, 2009) and high levels of self-confidence (Aydın & Erdem, 2022) contributed to academic resilience. Finally, it was found that resilient students had good advantages in physical aspects, including good eating and sleeping patterns, as well as few illnesses (Mandleco & Perry, 2000; Murphy, 1987).

2.3.2 Home context

Family support is a potential factor contributing to academic resilience. High levels of parenting supervision, family guidance, parental expectation, and a vigorous family environment play positive roles in students' academic resilience (Anagnostaki et al., 2016; García-Crespo et al., 2021; Li & Yeung, 2019; Li et al., 2017). In addition, a favorable child-parent interaction promoted disadvantaged students' academic achievement (Ma & Wu, 2019). Therefore, academically resilient students tend to have a supportive family atmosphere in face of challenging circumstances.

2.3.3 School context

School characteristics provide two crucial sources for nurturing academically resilient students. The first strand refers to school contextual factors. For example, individuals who study in a private school were more likely to become academically resilient students (Clavel et al., 2022; Vera et al., 2015). Furthermore, the availability of qualified teachers is a significant indicator for resilient schools (Agasisti & Longobardi, 2014). Of note, high-quantity and quality school resources are positively associated with academic resilience (Özberk et al., 2018; Vicente et al., 2021). While the second strand is more malleable, which pertains to school process factors. To be specific, teacher support acts as a protective role in academic resilience (Ferrera & Rodríguez, 2015; Özberk et al., 2018). Moreover, positive teacher morale and high-quality teaching practices increased the possibility of becoming ARS (Agasisti et al., 2021; Teng, 2020). Concerning the school climate, numerous studies revealed that an orderly and safe school provided a protective environment for fostering ARS (García-Crespo et al., 2021; Vicente et al., 2021). In other words, a better disciplinary climate, less disturbance and bullying problems in the classroom are positively associated with academic resilience (Aydın & Erdem, 2022; Cordero & Mateos-Romero, 2021).

2.4 The Asian context

Over the last few decades, resilience research on western-based populations is attracted great attention, while those findings might not be generalized to the Asian contexts (Li et al., 2017; Ungar, 2006). In Asia, although the percentages of poor populations have a rapid decline, over 40% of children still live in extreme poverty (World Bank, 2022). Moreover, migration, living in rural areas, gender discrimination, risky behaviors, and child labor bring huge barriers to Asian education systems (UNICEF, 2019). Despite these challenges, Asian economies [i.e., Beijing-Shanghai-Jiangsu-Zhejiang (B-S-J-Z), Hong Kong, Macao, Singapore, Taiwan, Japan, Korea] rank top in the league table of most ILSA (IEA, 2020; OECD, 2019a). It is worth noting that those high-performing Asian students might come from disadvantaged or advantaged families (OECD, 2010). In this sense, it is necessary to identify factors that contribute to Asian students' academic success, especially for disadvantaged individuals.

2.5 Conceptual framework and objective of the research

Ungar and Theron (2020) suggested that identifying dynamic processes within multiple systems led to a better understanding of resilience. Aligned with the social-ecological definition of resilience, key contextual features contain individual, family, community, as well as cultural contexts (Ungar, 2011). That is, resilience is a systemic concept examined at various levels of analysis. Borrowed from this core idea, an integrative conceptual framework of this study is developed based on previous literature and the PISA 2018 assessment framework (OECD, 2019b). Figure 1 depicts that three core domains contribute to academic resilience. First, student context contains cognitive skills, motivation, emotional styles, and behavioral engagement. Second, home context involves family resources, child-parent relationships, and parental practices. Finally, school basic information, teaching practices, school resources, and school climate are four categories under the school context.

The overarching objective of this study is to comprehensively identify key features (i.e., individual, home, and school contextual features) to discriminate ARS

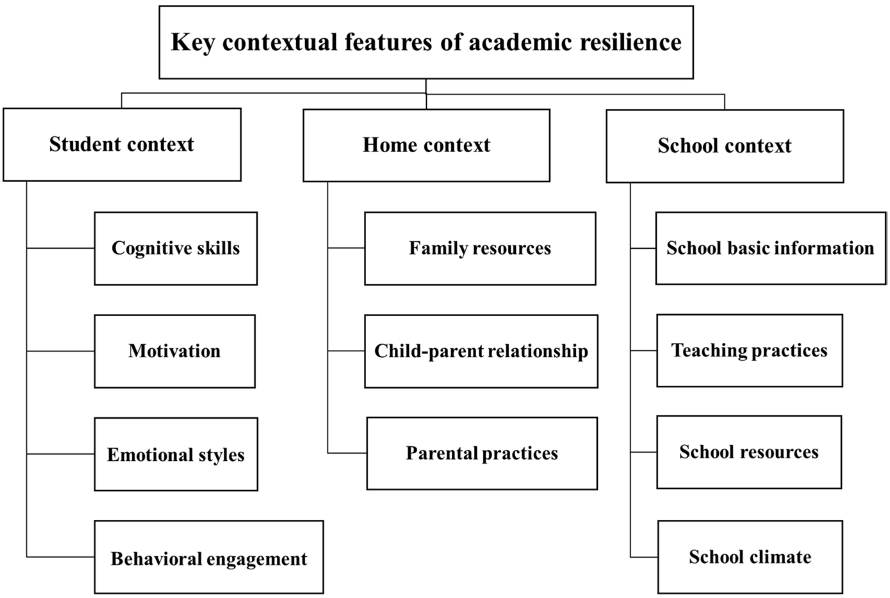

Fig. 1 An integrative conceptual framework of this study

from NRS. Our study provides new insights into academic resilience research in three ways. First, an extensive range of contextual features has been utilized in this study, which shows advantages over previous studies that only focused on limited factors (e.g., Agasisti et al., 2021; Cheung et al., 2014). Second, our study is dedicated to filling in the methodological gap in resilience research, by implementing an innovative ML approach rather than traditional statistical analysis (e.g., Li & Yeung, 2019; Ma & Wu, 2019). Finally, seven high-performing Asian economies are considered as our research sample. Rather than heavily emphasizing Western-based resilience research, the current study contributes to identify unique features through an Asian-specific lens.

3 Methods

3.1 Data

The data were drawn from the PISA 2018 database, which can be downloaded from the OECD's official website (https://www.oecd.org/pisa/data/2018database/). PISA has completed eight cycles of data collection (i.e., PISA 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2022), while the latest cycle has not released the results until now. In PISA 2018, reading is administered as a major domain, while science and mathematics are regarded as minor domains. Moreover, the assessment mode of PISA 2018 is computer-based testing.

A total of seven high-performing Asian economies were selected as our research sample, including B-S-J-Z, Hong Kong, Macao, Taiwan, Japan, Korea, and Singapore. Of these 48,548 fifteen-year-old participants in East Asia, we first screened the ESCS-disadvantaged students following the OECD's operational definition of ARS (OECD, 2019a). That is, students who are in the bottom 25% of ESCS within their economy are the major concern in our research. Second, within these targeted students, we further selected high performers whose digital reading performance achieved the top 25% across seven Asian economies. Overall, our final sample involved 11,496 students, which were categorized into 1,397 ARS and 10,099 NRS. Table 1 reveals the percentages of ESCS-disadvantaged students classified by ARS and NRS in each Asian economy.

3.2 Measures

PISA 2018 background questionnaires involve a wide range of non-cognitive constructs, such as student, teacher, parent, school, and ICT domains. For example, the item of *reading enjoyment* asked students to rate how much they agree or disagree with the statements about reading (i.e., "I read only if I have to"; "Reading is one of my favorite hobbies"; "I like talking about books with other people"; "For me, reading is a waste of time"; "I read only to get information that I need"), using a four-point Likert scale (i.e., strongly agree, agree, disagree, strongly disagree). Following the PISA 2018 assessment and analytical framework (OECD, 2019b), 105

Table 1 Percentages of ESCS-disadvantaged students by ARS and NRS classifications

	ARS (%)	NRS (%)	Total sample size (*N*)
B-S-J-Z	15.7	84.3	2,115
Hong Kong	16.7	83.3	1,713
Macao	15.5	84.5	941
Taiwan	6.2	93.8	1,846
Japan	5.8	94.2	1,559
Korea	10.1	89.9	1,623
Singapore	15.6	84.4	1,699
Total	12.2	88.8	11,496

ARS academically resilient students; *NRS* non-academically resilient students

contextual features derived from the background questionnaires have been utilized in this research. Selected independent variables were scaled by the item response theory, which were verified as high validity and reliability constructs in the PISA technical report (OECD, 2020). Of note, eligible features are those might have potential influences on academic resilience in previous literature.

The dependent variable was determined by two dimensions, referring to students who were at the bottom quarter of ESCS and achieved the top quarter of reading performance in PISA 2018. Due to the large sample size in PISA, no significant differences appear between one plausible value (PV) and the set of ten plausible values (PVs) (OECD, 2012). In this sense, we randomly chose the first PV of reading achievement (PVREAD1) to designate as students' performance. Hence, a binary dependent variable was developed to discriminate ARS from NRS.

3.3 Analytical strategy

3.3.1 Rationale for using support vector machine (SVM)

Traditional statistical analysis was broadly used to identify key features of academic resilience in most ILSA studies, such as multilevel logistic regression (Agasisti & Longobardi, 2014), structural equation modeling (Ma & Wu, 2019), and multiple regression (Li & Yeung, 2019). Nevertheless, those studies were mainly concerned with a limited group of features rather than a comprehensive cohort. This might be attributed to two main reasons. First, the multicollinearity issues and highly correlated effects caused by a mass of multilevel variables, which bring huge challenges for these conventional models to generate a flawless solution (Senaviratna & Cooray, 2019; Shieh & Fouladi, 2003). Second, the traditional methods tend to assume strong hypotheses, which primarily pay attention to a constrained set of features relying on in-depth theoretical grounding. Instead, our study does not make any robust preconditions, while an ML method is suitable for this situation. In line with our research design, a labeled output variable is coded by 1 (i.e., ARS) and 0 (i.e., NRS), and the input variables are also linked with the label. In this sense, a supervised ML method, rather than the unsupervised one, is more appropriate for our research (e.g., Chen et al., 2022).

Three supervised ML approaches prevalently applied to the ILSA research in education, including classification and regression trees (CART), random forest (RF), and support vector machine (SVM) (e.g., Chen et al., 2022; Kılıç Depren & Depren, 2022; Yi & Na, 2020). However, CART is easily influenced by slight changes (Kuhn, 2008) and suffered from overfitting issues (Chen & Guestrin, 2016). Although overfitting might be mitigated by RF, it still confronts the concern of inconsistent forests caused by over-subsampling, and entirely randomized trees could also result in overfitting (Tang et al., 2018). Unlike CART and RF, SVM avoids the above problems because it does not directly dispose of the high dimensional space (Boswell, 2002). Compared with conventional classification techniques, SVM appears to be superior in terms of performance balance (Quan & Pu, 2022). In addition, SVM is particularly efficient to deal with high-order dimensional data (Chen et al., 2022), has an accurate and robust ability in classification (Shao & Lunetta, 2012), and generate a unique solution if it has rounded optimality (Karamizadeh et al., 2014). Well-documented by previous ILSA studies, SVM with recursive feature elimination (RFE) has been proven its excellence to identify a massive group of features in the classification model (e.g., Chen et al., 2021a, b; Hu et al., 2022a, b).

3.3.2 Support vector machine-recursive feature elimination (SVM-RFE)

SVM is a binary classifier in search of the optimal margin with kernel functions to divide support vectors into two groups in a multidimensional hyperplane (Cortes & Vapnik, 1995). To further acquire the key features in the classification of ARS and NRS, the algorithm of RFE is combined with SVM. The rationale for adopting RFE is two-fold. First, SVM is unable to process feature selection without an extension of RFE (Chen et al., 2022). While SVM-RFE is conducive to determine eligible feature subsets and has a high classification accuracy of the model (Huang et al., 2014). Second, SVM-RFE is much easier to avoid overfitting problems and its processing time is faster than other feature selection methods (Gorostiaga & Rojo-Álvarez, 2016; Guyon et al., 2002). Overall, there are three steps in the iterative mechanism of SVM-RFE. First, the weight vectors are calculated after an optimal model is acquired. Second, a ranking coefficient of the feature is assigned and the smallest one is eliminated at each iteration. In this sense, this process recalculates the feature relevance and reduces the number of features each time. Finally, a list of optimal features ranking in descending order is presented (Guyon et al., 2002).

3.4 Analytical procedures

The data analysis followed a three-stage procedure, including a four-step data preprocessing, a two-step primary analysis, and the model evaluation. Python 3.9.13 program was utilized to implement all data analysis procedures. In addition, it is publicly free access to the coding documentation libraries from the website of *scikit-learn* (https://scikit-learn.org/stable/) and *imbalanced-learn* (https://imbalanced-learn.org/dev/).

Data preprocessing The first step of data preprocessing is data cleaning. A missing or invalid response was identified as the noise (e.g., ‘response = 999’ in a four-point Likert scale item) in the initial dataset. Second, the *k*-nearest neighbor (kNN) imputation was adopted to impute the noise data. Third, we further conducted the normalization to standardize the data ranging from 0 to 1. Finally, a combined sampling strategy (i.e., SMOTETomek) was applied to resample the imbalanced dataset since there was an extreme percentile gap between the ARS and NRS classification.

Primary analysis Beforehand, the pre-processed dataset was split into the test set (20%) and the training set (80%). For the training set, we first implemented an SVM model with a linear kernel function. Due to an imbalanced gap between the minority and majority classes (i.e., ARS vs. NRS), different weights were given to them with the class weight function in SVM. Subsequently, SVM-RFE was further conducted to select and rank the key features in the identification of two individual cohorts. In the last phase, we applied a ten-fold cross-validation to determine the optimal subset of key features derived from 105 contextual features. That is, nine folds (i.e., training set) were randomly chosen for model estimation, while the remaining fold (i.e., test set) was assigned as model validation.

Model evaluation There is no certain principle to determine the optimal number of *k* in k-folds cross-validation, while it is common practice to use between 5 and 10 (Arlot & Celisse, 2010). Moreover, it might be preferable to use 10 since a larger *k* could result in a smaller bias (Kuhn & Johnson, 2013). Therefore, all model evaluation metrics were acquired through ten-fold cross-validation, including accuracy, the area under the receiver operating characteristic curve (AUC), recall, precision, and F-measure. To be specific, each metric is elucidated by its mathematical formula below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$AUC = \int_0^1 ROC(t)dt \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$F - measure = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

Four common indicators (i.e., TP, TN, FP, FN) in these formulae are explained as follows: (1) TP refers to the number of ARS correctly predicted; (2) TN refers to the number of NRS correctly predicted; (3) FP refers to the number of ARS predicted as NRS, and (4) FN refers to the number of NRS predicted as ARS. Overall, accuracy is widely utilized to evaluate the classification performance of a model. Since it is not enough to be merely evaluated by the accuracy score in practical applications, AUC has been further proven to be more discriminating and statistically consistent than accuracy (Huang & Ling, 2005). Especially, accuracy is commonly ineffective if the dataset is imbalanced. Hence, it is advised that recall and precision are more applicable in the classification of imbalanced samples (Juba & Le, 2019). In addition, as a harmonic average of recall and precision, F-measure offers more enlightenment than accuracy to the classified function of a model with an imbalanced dataset (Bekkar et al., 2013). Therefore, it is suggested to consider various metrics for the SVM model evaluation, especially in our imbalanced cases of seven Asian economies.

4 Results

4.1 Prediction performance of the SVM model

By implementing ten-fold cross-validation, accuracy, AUC, recall, precision, and F-measure were calculated ten times. Table 2 shows these five metrics and their average scores at each fold, with a high average accuracy of 87.51% (AUC = 94.86%, Recall = 95.98%, Precision = 83.11%, F-measure = 88.63%). In general, an indicator scores greater than 80% is considered as good effectiveness (Xia et al., 2013). In this sense, the SVM model has a high prediction performance.

According to Gorostiaga and Rojo-Álvarez (2016), the minimum number of optimal features refers to 20 to 30 in the analysis of PISA studies through SVM. To determine the number of optimal feature sets, the accuracy of top 35 feature sets was

Table 2 Prediction performance (%) of the SVM model by ten-fold cross-validation in classification of academic resilience

k-fold	Accuracy	AUC	Recall	Precision	F-measure
1	86.68	94.97	78.32	94.05	85.47
2	92.03	97.08	93.07	91.17	92.11
3	84.80	93.97	99.50	76.89	86.75
4	92.67	98.03	97.92	88.62	93.04
5	93.86	98.78	97.82	90.64	94.10
6	90.54	97.47	98.41	85.03	91.23
7	76.08	84.14	98.81	67.92	80.50
8	86.13	94.27	98.81	78.81	87.69
9	87.32	95.22	98.81	80.35	88.63
10	84.99	94.64	98.32	77.64	86.76
Average	87.51	94.86	95.98	83.11	88.63

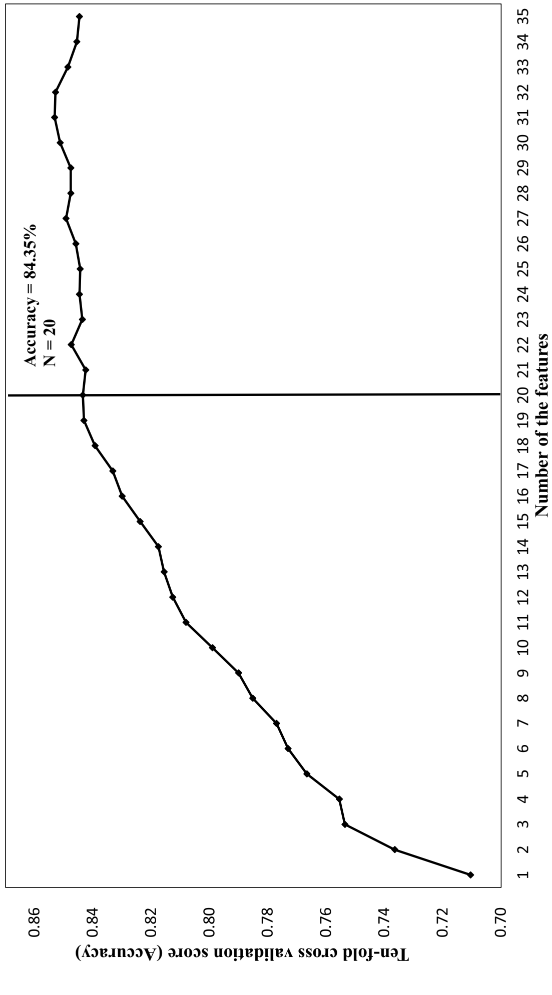

Fig. 2 The accuracy of top 35 feature sets with ten-fold cross-validation

calculated with ten-fold cross-validation (Fig. 2). There was a slight difference among the top 20 (84.35%), top 25 (84.44%), top 30 (85.13%), and top 35 (84.46%) feature sets. Therefore, after the 20th feature, the accuracy tended to remain relatively convergent. This tendency is consistent with Hu et al.'s (2022a) findings. Hence, the top 20 feature set was regarded as the optimal combination. This result is also confirmed by previous ILSA research that adopted SVM (Chen et al., 2021b, 2022).

Table 3 shows the prediction performance of the top 20 features by ten-fold cross-validation, with a high average accuracy of 84.35% (AUC = 92.34%, Recall = 91.51%, Precision = 80.69%, F-measure = 85.42%). Thus, the metrics of the top 20 features exhibited a high similarity of classification performance with the complete 105 features. It is further validated that the top 20 contextual features are of great significance in the classification of ARS and NRS.

4.2 Descriptions of the top 20 key features

The top 20 key features to discriminate ARS from NRS are presented in the descending importance of classification performance (Table 4). Detailed descriptive analysis of these top features is also elucidated. ARS generally showed high levels of metacognitive strategies in reading, including assessing credibility (METASPAM), summarizing (METASUM), as well as understanding and remembering (UNDREM). In addition, ARS was less likely to repeat a grade (REPEAT) at school. Regarding the affective features, reading enjoyment (JOYREAD), perception of the difficulty in PISA test (PISADIFF), student's expected occupational status (BSMJ), feelings of meaning in life (EUDMO), and sense of belonging (BELONG) also played crucial roles in the classification of ARS and NRS. One interesting finding is that the ICT-related features show controversial impacts. Compared with NRS, ARS tended to have more ICT resources at home (ICTRES) and autonomy in ICT use (AUTICT), while less consideration on ICT as a topic in social interaction (SOIAICT). Moreover, ARS was more likely to have more home educational resources (HEDRES). With respect to the teaching practices, feedback provided by the

Table 3 Prediction performance (%) of the top 20 features by ten-fold cross-validation in classification of academic resilience

k-fold	Accuracy	AUC	Recall	Precision	F-measure
1	82.28	91.82	73.37	89.28	80.54
2	88.51	94.58	90.59	86.98	88.75
3	80.54	89.72	93.76	74.16	82.82
4	87.67	95.36	93.56	83.70	88.36
5	89.01	95.68	93.07	86.08	89.44
6	89.06	95.38	93.56	85.83	89.53
7	80.88	89.56	94.65	74.20	83.19
8	83.61	91.42	94.55	77.56	85.22
9	81.43	90.57	94.36	74.98	83.56
10	80.49	89.28	93.66	74.14	82.76
Average	84.35	92.34	91.51	80.69	85.42

Table 4 A description of the top 20 features in the classification of ARS and NRS

Rank	*Context*	*Variable*	*Description*	*Overall M (SD)*	*ARS M (SD)*	*NRS M (SD)*
1	Student	METASPAM	Meta-cognition: assess credibility	–0.15 (1.02)	0.56 (0.83)	–0.23 (1.01)
2	Student	METASUM	Meta-cognition: summarizing	–0.30 (1.02)	0.41 (0.76)	–0.39 (1.02)
3	Student	JOYREAD	Joy/Like reading	0.33 (0.97)	0.88 (0.85)	0.26 (0.96)
4	Student	REPEAT	Grade Repetition	0.11 (0.32)	0.06 (0.25)	0.12 (0.33)
5	Student	PISADIFF	Perception of difficulty of the PISA test	0.45 (0.99)	–0.03 (0.86)	0.51 (0.99)
6	Student	BSMJ	Student's expected occupational status	57.50 (20.61)	67.33 (18.12)	56.18 (20.56)
7	Student	UNDREM	Meta-cognition: understanding and remembering	–0.06 (1.04)	0.50 (0.83)	–0.13 (1.05)
8	Home	ICTRES	ICT resources at home	–1.06 (0.71)	–0.95 (0.65)	–1.07 (0.71)
9	School	STUBEHA	Student behaviour hindering learning	0.05 (1.50)	–0.34 (1.82)	0.10 (1.45)
10	Student	EUDMO	Eudaemonia: meaning in life	–0.19 (0.94)	–0.32 (0.93)	–0.18 (0.95)
11	School	TCDISCLIMA	Disciplinary climate in test language lessons perceived by teachers	0.23 (0.55)	0.45 (0.41)	0.20 (0.56)
12	Home	HEDRES	Home educational resources	–0.80 (0.97)	–0.51 (0.99)	–0.84 (0.96)
13	Student	BELONG	Subjective well-being: Sense of belonging to school	–0.13 (0.88)	–0.22 (0.81)	–0.12 (0.89)
14	Teacher	FEEDBACK	Feedback provided by the teachers	–0.11 (0.31)	–0.17 (0.32)	–0.10 (0.31)
15	Student	SOIAICT	ICT attitude: ICT as a topic in social interaction	–0.32 (1.07)	–0.41 (0.92)	–0.32 (1.08)
16	Student	AUTICT	ICT attitude: Perceived autonomy related to ICT use	–0.25 (1.05)	0.00 (0.92)	–0.27 (1.06)
17	School	DISCLIMA	Disciplinary climate in test language lessons perceived by students	0.60 (1.06)	0.80 (0.96)	0.58 (1.06)
18	Teacher	DIRINS	Teacher-directed instruction	0.26 (1.05)	0.24 (0.90)	0.26 (1.07)
19	School	PERCOMP	Perception of competitiveness at school	–0.11 (1.01)	0.08 (0.95)	–0.14 (1.02)
20	School	TOTAT	Total number of all teachers at school	100.62 (85.27)	133.53 (93.42)	96.66 (83.36)

ARS academically resilient students; *NRS* non-academically resilient students teachers (FEEDBACK) and teacher-directed instruction (DIRINS) were crucial features in discriminating ARS from NRS. Regarding the school climate and its basic background, student behavior hindering learning (STUBEHA), disciplinary climate (TCDISCLIMA and DISCLIMA), competitive atmosphere at school (PERCOMP), as well as total number of teachers at school (TOTAT) could differentiate between the two student cohorts.

5 Discussion

Following the conceptual framework of our study, the top 20 features were reorganized to have an in-depth discussion of the key findings (Table 5). Given that personal experience took up eleven out of twenty features, it played a predominant role in discriminating ARS from NRS. Especially, the use of metacognitive strategies in digital reading and enjoyment of reading were the most influential features of academic resilience. Likewise, it is evidenced that metacognitive awareness of reading strategies fostered resilient students in Asia (e.g., Cheung et al., 2014; Liu & Zhai, 2020). In addition, reading enjoyment is well-documented as a key protective factor of academic resilience (e.g., Özcan & Bulus, 2022; Vicente et al., 2021). Rooted in the self-determination theory, reading enjoyment is regarded as intrinsic motivation (Ryan & Deci, 2000). Interestingly, a recent study indicated that the use of metacognition strategies mediated the association between reading motivation and reading performance of disadvantaged students (Jang et al., 2023). In other words, individuals who autonomously transformed their reading motivation into reading strategies are more likely to become ARS. Thus, ARS not only intrinsically enjoy reading but also monitor their ways of thinking and action.

Compared with the individual factors, family context was a less significant domain in this study. The little influence might attribute to the low responses of parent questionnaires in most sampled Asia economies. Of note, ICT resources at home and home educational resources are two crucial features in the classification of ARS and NRS. The index of ICT resources at home incorporates a set of measures, for example, the availability of desktop computers, tablet computers, cellphones, and internet connection at home. In addition, home educational resources also contain ICT-related items, including educational software and computer use for schoolwork at home. Thus, the influence of ICT on academic resilience becomes the major focus of our discussion.

The result showed that resilient students had more opportunities to access ICT resources at home. It seems the usability of ICT resources contributes to nurturing resilient students. However, most studies indicated that the number of computers at home and academic resilience were not significantly related (Ferrera & Rodríguez, 2015; Marcenaro, 2014). The main reasons behind these contradictory results are two-fold. First, there is no causal relationship in the cross-sectional research design. It is inappropriate to infer the cause-and-effect conclusion between ICT resources and academic resilience. Second, rather than the availability of computers, the fundamental factor of academic resilience might be how students perceive and utilize

Table 5 A description of the top 20 features classified by different domains

Domain	*Sub-domain*	*Rank*	*Variable*	*Description*
Personal experience	Cognitive skills	1	METASPAM	Meta-cognition: assess credibility
		2	METASUM	Meta-cognition: summarizing
		7	UNDREM	Meta-cognition: understanding and remembering
	Motivation	3	JOYREAD	Joy/Like reading
		5	PISADIFF	Perception of difficulty of the PISA test
		6	BSMJ	Student's expected occupational status
		15	SOIAICT	ICT attitude: ICT as a topic in social interaction
		16	AUTICT	ICT attitude: Perceived autonomy related to ICT use
	Behavioral engagement	4	REPEAT	Grade Repetition
	Emotional styles	10	EUDMO	Eudaemonia: meaning in life
		13	BELONG	Subjective well-being: Sense of belonging to school
Family support and atmosphere	Family resources	8	ICTRES	ICT resources at home
		12	HEDRES	Home educational resources
School contextual and process factors	School climate	9	STUBEHA	Student behaviour hindering learning
		11	TCDISCLIMA	Disciplinary climate in test language lessons perceived by teachers
		17	DISCLIMA	Disciplinary climate in test language lessons perceived by students
		19	PERCOMP	Perception of competitiveness at school
	Teaching practices	14	FEEDBACK	Feedback provided by the teachers
		18	DIRINS	Teacher-directed instruction
	School basic information	20	TOTAT	Total number of all teachers at school

ICT. Our findings revealed that ARS was less likely to use ICT in social interaction, which is consistent with Clavel et al. (2022). It might interpret that ARS tended to make friends and exchange their ideas through face-to-face talk. Furthermore, our results indicated that ARS tended to perceive more autonomy related to ICT use. Perceptions towards autonomy use in ICT refer to the feelings of being independent and responsible for technology (Courtney et al., 2022). Of note, autonomy is one of the basic psychological needs based on the self-determination theory (Ryan & Deci, 2017). There was a positive relationship between autonomy needs and intrinsic motivation (Mouratidis et al., 2018; Vasquez et al., 2016). Therefore, the significant role of motivation in academic resilience might contribute to explaining why ARS tend to satisfy their autonomous need in ICT-related activities (Martin & Marsh, 2006). It thus infers that ARS is inclined to independently read information about technology and solve technical problems with digital devices on their own.

Regarding the school contexts, we would like to highlight the crucial impact of school climate on academic resilience. Our results indicated that ARS was more likely to study in an orderly and structured class. In accordance with this finding, prior research evidenced that disciplinary climate buffered the negative impact of disadvantaged backgrounds on students' academic resilience (Teng, 2020). In other words, a favorable disciplinary climate is conducive to promoting resilient students (Agasisti et al., 2021; Vicente et al., 2021). One striking finding in our study is that ARS tends to learn in a competitive-driven schooling environment. This also accords with Özcan and Bulus's (2022) work. Interestingly, they further clarified that only collectivistic countries depicted this outcome whereas individualistic countries did not. It thus evidenced the cultural influence on the construction of academic resilience (Ungar, 2011). Therefore, it is worthwhile to further investigate if the competitive atmosphere at school becomes a unique feature of academic resilience in the collectivistic culture, especially for Asian economies.

6 Theoretical and practical implications

The current research offers fresh insights into the development of academic resilience research in several ways. Concerning the theoretical implications, a massive group of features was considered in the integrative framework. Thus, it addresses the limitation that prior research only paid attention to limited factors in academic resilience. Moreover, an innovative ML technique was implemented in this study. It fills a methodological gap because most studies adopted conventional statistical models to identify the protective factors of academic resilience. Since there is an extreme imbalance between the number of ARS and NRS, our study explicitly exemplifies how to deal with the imbalanced dataset through the resampling and weighting strategies through an ML approach. Furthermore, our study also niches an evidence gap because few resilience studies focus on the Asia context, especially in ILSA research.

With respect to practical insights, students should be taught various metacognition strategies in digital reading (e.g., assessing credibility, summarizing, understanding, and remembering). Therefore, teachers need to provide regular practices for children to apply and reflect on these strategies. To motivate students' reading engagement, teachers and parents could provide more opportunities for children to read, offer advice to choose different reading materials, and encourage them to read independently (Curriculum Development Council, 2017). Concerning the impact of ICT on academic resilience, students are encouraged to communicate with their friends 'physically', instead of immersing themselves in a virtual world. Moreover, autonomous-supportive teaching practices and parenting styles play vital roles in satisfying students' autonomous needs in ICT (Areepattamannil & Santos, 2019). An intervention program has suggested six instructional methods to develop teachers' empowering and encouraging style, including 1) showing patience; 2) considering student's point of view; 3) using invitational words; 4) activating the psychological needs of students; 5) giving justifications for teacher's demands; and 6) recognizing and accepting negative emotional manifestations (Cheon et al., 2018). Following a parenting intervention program, it is recommended for parents to resonate with children and emphasize the meaningful component of school assignments without heavy control (e.g., bringing pressure or using external rewards) (Froiland, 2015). Finally, for nurturing resilient students in Asia, school administrators need to maintain a healthy competitive atmosphere on the campus. In addition, rational, firm, and clear school regulations should be set to guarantee an orderly disciplinary climate.

7 Limitations and directions for future research

Although this study provides new insights into theoretical and practical implications in education, several limitations should be addressed for future studies. First, limited by the characteristics of ML approach, it is challenging for SVM to interpret the interaction effects in multiple key features. Second, PISA implements a cross-sectional research design, which could not infer a causal relationship between the key features and academic resilience. Finally, only seven high-performing Asian economies were considered as our research sample. In this sense, the implications for the results could not generalize to the Western context. Since the targeted population is fifteen-year-old students in PISA, our results are constrained to the stage of secondary education. Therefore, future work should adopt traditional statistical techniques to deeply examine the possible non-linear or hierarchical relationships between key features and academic resilience. Moreover, longitudinal ILSA should be considered as the data source to confirm the validity of the top 20 features we selected from SVM. Considering the generalizability of research findings, students from Western economies and different educational phases should be selected as research samples in the future.

8 Conclusion

The current study contributes to identifying key features of academic resilience in PISA 2018. Through an ML approach, the top 20 features were selected to discriminate ARS from NRS. Overall, metacognition strategies in digital reading and reading enjoyment were predominant features in this study. These findings fill the evidence and methodological gaps, which provide fresh insights for academic resilience research in the online learning environment.

Author's contribution JQ: writes and revises the manuscript. KC and PS: supervise, revise, and proofread the manuscript. All authors contributed to the article and approved the submitted version.

Data availability Publicly available datasets were analyzed in this study. The data that support the findings of this study are openly available at https://www.oecd.org/pisa/data/2018database/.

Declarations

Statement regarding informed consent/ethical approval/ research involving human participants and/or animals This study involved analysis of secondary data that is publicly available, and the informed consent/ethical approval/research involving human participants and/or animals are not applicable here.

Competing interests On behalf of all authors, the corresponding author states that there is no conflict of interest.

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